Mathematics practice booklet: Grade 7 expectations

Grade 7 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 7 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Draft: to be checked by a teacher before use.

Section 1: Patterns, equations and algebra thinking

Standards practised: 7.EE.A.1, 7.EE.A.2, 7.EE.B.3, 7.EE.B.4

A worked example

Solve 3x + 5 = 20. Then check your answer.

1. Undo the + 5 by subtracting 5 from both sides: 3x = 15.
2. Undo the multiplication by dividing both sides by 3: x = 5.
3. Check: 3 x 5 + 5 = 15 + 5 = 20, which is correct.

x = 5

Questions

1. Simplify: 5x + 3 + 2x - 1

2. Expand: 4(x + 3)

3. Factor: 6x + 9

4. Expand and simplify: 2(3x - 4) + 5

5. Simplify: (5x - 2) - (2x - 6)

6. Solve: 2x + 7 = 19

7. Solve: 3(x - 2) = 15

8. Solve: x / 4 - 3 = 5

9. Solve the inequality: $2x + 3 > 11$

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10. Salem buys 3 cups of karak and one pastry that costs AED 4. He pays AED 19 in all. Write an equation and find the price of one cup.

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11. Maryam has AED 40 on her metro card. Each trip costs AED 3. Write an inequality for the number of trips n she can take and say the greatest whole number of trips.

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12. Hamdan says $3(x + 2)$ is the same as $3x + 2$. Is he right? Explain.

Section 2: Shapes, measuring and space

Standards practised: 7.G.A.1, 7.G.A.2, 7.G.B.4, 7.G.B.5, 7.G.B.6

A worked example

A circular fountain has a radius of 7 m. Using pi = 22/7, find its circumference and its area.

1. Circumference = 2 x pi x r = 2 x 22/7 x 7 = 44 m.
2. Area = pi x r x r = 22/7 x 7 x 7 = 22 x 7 = 154.
3. Area is measured in square units: 154 square metres.

Circumference 44 m, area 154 square metres

Questions

1. On a map, 1 cm stands for 5 km. Two towns are 6 cm apart on the map. What is the real distance?

2. A room is 4 m by 3 m. In a scale drawing, 1 cm stands for 0.5 m. What are the sides of the room in the drawing?

3. Can a triangle be drawn with sides 3 cm, 4 cm and 8 cm? Explain.

4. A circle has a diameter of 10 cm. Find its circumference. Use pi = 3.14.

5. A circle has a radius of 5 cm. Find its area. Use pi = 3.14.

6. A circular table has a circumference of 44 cm. Using pi = 22/7, what is its diameter?

7. Two angles on a straight line are next to each other. One is 115 degrees. Find the other.

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8. Two straight lines cross. One of the angles is 50 degrees. Find the angle opposite it and the angle next to it.

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9. A floor is a rectangle 10 m by 6 m with a rectangular corner 4 m by 3 m cut out. What is the area of the floor?

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10. A triangular prism has a triangular end with base 6 cm and height 4 cm. The prism is 10 cm long. Find its volume.

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11. Find the surface area of a box that is 5 cm by 4 cm by 3 cm.

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12. A water tank is 2 m long, 1.5 m wide and 1 m high. How many litres does it hold when full? (1 cubic metre = 1,000 litres)

Section 3: Data, graphs and chance

Standards practised: 7.SP.A.1, 7.SP.A.2, 7.SP.B.3, 7.SP.B.4, 7.SP.C.5, 7.SP.C.6, 7.SP.C.7, 7.SP.C.8

A worked example

A bag holds 2 red, 3 green and 5 yellow beads. One bead is taken without looking. What is the probability that it is green?

1. Count all the beads: 2 + 3 + 5 = 10.
2. Green beads: 3.
3. Probability = 3 out of 10 = 3/10 = 0.3.

3/10

Questions

1. What does a probability of 0.9 tell you about an event?

2. A bag has 4 red marbles and 6 blue marbles. One is picked without looking. What is the probability it is blue?

3. A fair coin is tossed twice. What is the probability of two heads?

4. A fair coin is tossed once and a spinner with 3 equal colours (red, green, blue) is spun once. How many different outcomes are possible?

5. Layla spins a spinner 50 times and it lands on green 18 times. What is the experimental probability of green?

6. The probability of landing on green is 1/4. About how many times would you expect green in 80 spins?

7. A school wants to know the favourite fruit of its 500 students. Which sample is better: 30 students chosen at random from the school list, or 30 students waiting at the fruit stall? Why?

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8. In a random sample of 40 students, 10 prefer football. Predict how many of the 600 students in the school prefer football.

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9. Team A has a mean height of 150 cm and Team B has a mean height of 156 cm. Both groups have a mean absolute deviation of 4 cm. The difference between the means is how many times the mean absolute deviation?

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10. Group A scores: 5, 6, 7, 8, 9. Group B scores: 8, 9, 10, 11, 12. How much higher is the median of group B?

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11. A seed sprouts with a probability of 40%. You use random digits from 0 to 9 to simulate one seed. Which digits could stand for sprouting?

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12. Hamdan tosses a fair coin and gets heads 4 times in a row. He says tails is now more likely. Is he right? Explain.

Section 4: Numbers and calculating

Standards practised: 7.RP.A.1, 7.RP.A.2, 7.RP.A.3, 7.NS.A.1, 7.NS.A.2, 7.NS.A.3

A worked example

A bag costs AED 80 before VAT. VAT is 5%. What is the price with VAT?

1. VAT = 5% of 80 = 0.05 x 80 = AED 4.
2. Price with VAT = 80 + 4 = AED 84.
3. Another way: multiply by 1.05: 80 x 1.05 = 84.

AED 84

Questions

1. A boy walks 1/2 km in 1/4 of an hour. What is his speed in km per hour?

2. 3 kg of dates cost AED 54 at the souq. At the same price, what do 5 kg cost?

3. A table pairs x and y: (2, 6), (4, 12), (5, 16). Is y proportional to x? Explain.

4. A shirt costs AED 120. It is sold at 25% off. What is the sale price?

5. A pair of shoes costs AED 240 before VAT. What is the total with 5% VAT?

6. A trader at the souq buys a rug for AED 40 and marks it up by 30%. What is the selling price?

7. Noura saves AED 2,000 at 4% simple interest a year. How much interest does she earn in 3 years?

8. The price of a ticket rises from AED 50 to AED 60. What is the percent increase?

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9. A diver is 12 m below the sea surface (-12 m). She swims up 5 m. Where is she now?

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10. Work out: (a) -4 - (-9) (b) (-6) x (-7)

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11. Work out: -3/4 + 1/2

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12. A jacket costs AED 150. It is reduced by 20%, and then 5% VAT is added to the reduced price. What does Omar pay? Show the steps.

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. 7x + 2 (5x + 2x = 7x and 3 - 1 = 2)
2. 4x + 12
3. 3(2x + 3) (3 is a common factor of 6 and 9)
4. 6x - 3 (6x - 8 + 5)
5. 3x + 4 (subtracting -6 adds 6)
6. x = 6
7. x = 7 (x - 2 = 5, so x = 7)
8. x = 32 (x / 4 = 8)
9. x > 4
10. 3c + 4 = 19, so c = 5: a cup costs AED 5.
11. 3n ≤ 40, so n ≤ 13.3...; she can take at most 13 trips.
12. No. The 3 multiplies both terms: 3(x + 2) = 3x + 6.

Section 2: Shapes, measuring and space

1. 30 km
2. 8 cm by 6 cm
3. No: 3 + 4 = 7 is less than 8, so the two short sides cannot reach each other.
4. 31.4 cm (3.14 x 10)
5. 78.5 square cm (3.14 x 5 x 5)
6. 14 cm (44 divided by 22/7)
7. 65 degrees (the two add to 180)
8. Opposite: 50 degrees. Next to it: 130 degrees (180 - 50).
9. 48 square metres (60 - 12)
10. 120 cubic cm (area of the end is 12, times 10)
11. 94 square cm (2 x (20 + 15 + 12))
12. 3,000 litres (2 x 1.5 x 1 = 3 cubic metres)

Section 3: Data, graphs and chance

1. It is very likely to happen (close to 1, which means certain).
2. 6/10 = 3/5
3. 1/4 (the outcomes are HH, HT, TH, TT, so 1 out of 4)
4. 6 (2 x 3)
5. 18/50 = 9/25 = 0.36
6. About 20 (1/4 of 80)
7. The random list: every student has a fair chance, so it is more likely to represent the whole school.
8. About 150 (10/40 = 1/4, and 1/4 of 600)
9. 1.5 times (6 / 4)
10. 3 (the medians are 7 and 10)
11. Any 4 of the 10 digits, for example 0, 1, 2, 3 (4 out of 10 is 40%).
12. No. Each toss is independent, so the chance of tails is still 1/2.

Section 4: Numbers and calculating

1. 2 km per hour (1/2 divided by 1/4)
2. AED 90 (AED 18 for each kg)
3. No: 6/2 = 3 and 12/4 = 3 but 16/5 = 3.2, so the ratio is not always the same.
4. AED 90 (discount 30, 120 - 30)
5. AED 252 (VAT is 12)
6. AED 52 (markup 12)
7. AED 240 (80 each year)
8. 20% (increase 10, and 10/50 = 0.2)
9. -7 m (7 m below the surface)
10. (a) 5 (subtracting -9 is adding 9) (b) 42 (two negatives make a positive)
11. -1/4 (-3/4 + 2/4)
12. AED 126 (150 x 0.8 = 120, then 120 x 1.05 = 126)